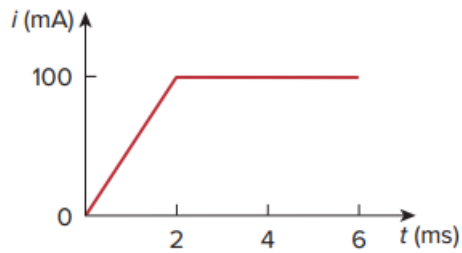


Problem

An initially uncharged 1-mF capacitor has the current shown in Fig. 6.11 across it. Calculate the voltage across it at $t = 2$ ms and $t = 5$ ms.

Fig. 6.11:

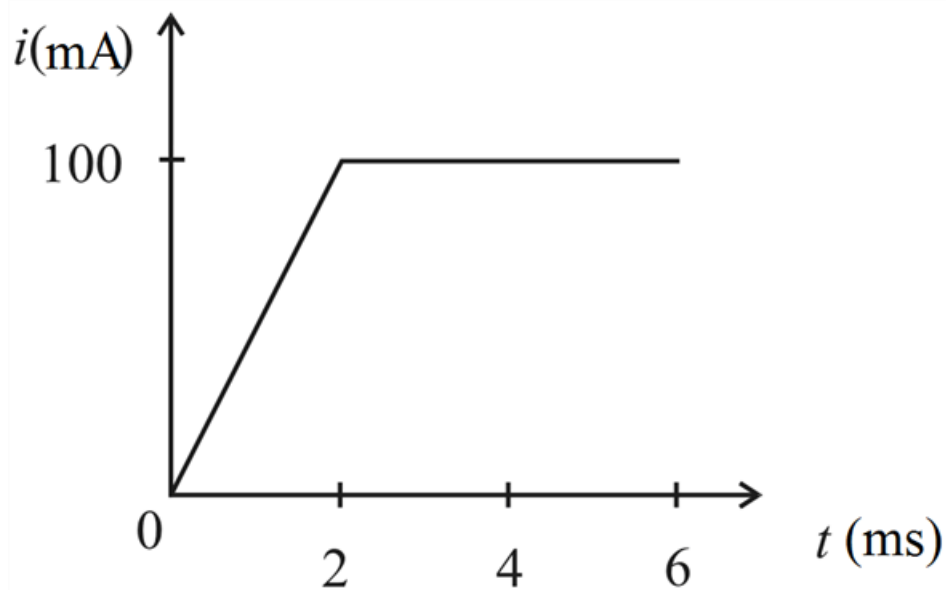


Step-by-step solution

Step 1 of 4

Given capacitor $C = 1\text{mF}$

Current source across the capacitor is



Step 2 of 4

Consider interval $0 < t < 2$ ms

In this interval two points on the straight line are $(0,0)$ and $(2\text{m},100\text{m})$

Line equation for these points is

$$i - 0 = \left(\frac{100 \times 10^{-3} - 0}{2 \times 10^{-3} - 0} \right) (t - 0)$$

$$i = (50)(t)$$

$$i = 50t \text{ A}$$

Consider interval $t > 2\text{ms}$

$$i = 100\text{mA}$$

Therefore we can write

$$i(t) = \begin{cases} 50t \text{ A} & \text{for } 0 < t < 2\text{ms} \\ 100 \text{ mA} & \text{for } t > 2\text{ms} \end{cases}$$

For $0 < t < 2 \text{ ms}$ voltage across capacitor is,

$$v = \frac{1}{C} \int_0^t i dt + v(0)$$

$$v = \frac{1}{1 \times 10^{-3}} \int_0^t (50t) dt + 0 \quad (\because v(0) = 0\text{V})$$

$$v = \frac{1}{1 \times 10^{-3}} \left(50 \left[\frac{t^2}{2} \right]_0^t \right)$$

$$v = \frac{50}{1 \times 10^{-3}} \left(\frac{t^2 - 0}{2} \right)$$

$$v = \frac{50}{1 \times 10^{-3}} \left(\frac{t^2}{2} \right)$$

Therefore voltage across capacitor at $t = 2\text{ms}$ is

$$v = \frac{50}{1 \times 10^{-3}} \left(\frac{(2 \times 10^{-3})^2}{2} \right)$$

$$v = 0.1$$

$$\boxed{v = 100\text{mV}}$$

Step 4 of 4

For $t > 2 \text{ ms}$ voltage across capacitor is,

$$v = \frac{1}{C} \int_0^t i dt + v(2\text{ms}) \quad \text{Here } v(2\text{ms}) \text{ is initial voltage}$$

$$v = \frac{1}{1 \times 10^{-3}} \int_{2\text{m}}^t (100 \times 10^{-3}) dt + 100 \times 10^{-3}$$

$$v = \frac{100 \times 10^{-3}}{1 \times 10^{-3}} [t]_{2\text{m}}^t + 100 \times 10^{-3}$$

$$v = \frac{100 \times 10^{-3}}{1 \times 10^{-3}} (t - 2 \times 10^{-3}) + 100 \times 10^{-3}$$

Therefore voltage across capacitor at $t = 5\text{ms}$ is

$$v = \frac{100 \times 10^{-3}}{1 \times 10^{-3}} (5 \times 10^{-3} - 2 \times 10^{-3}) + 100 \times 10^{-3}$$

$$v = 100 (3 \times 10^{-3}) + 100 \times 10^{-3}$$

$$v = 300 \times 10^{-3} + 100 \times 10^{-3}$$

$$v = 400 \times 10^{-3}$$

$$\boxed{v = 400\text{mV}}$$